

# ReTRACE: MODEL-RESPONSE WATERMARKING FOR DIFFUSION LANGUAGE MODELS

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## ABSTRACT

Existing language-model watermarks are detected from token patterns. We detect ours from *how a diffusion language model responds when the generated text is partially masked and reconstructed*. Because such a model is trained to recover masked tokens from arbitrary surviving context, a finished text can be pushed back through the operator that produced it: corrupted with a secret key, re-denoised, and its response read. ReTrace places a watermark in that response. A keyed pair of context ablations induces a per-position log-likelihood contrast, and the watermark asks the sign of that contrast to agree with a keyed orientation, giving a test statistic that is a sum of bounded indicators with a finite-sample exact  $\text{Binomial}(n, \frac{1}{2})$  null. Embedding never overrides the model: decoding follows the reference block schedule, and the watermark only decides whether a sampled proposal is committed or resampled. We also report the negative result that motivated this design — post-hoc token substitution is priced out on a well-trained diffusion language model, and, because both are governed by the sharpness of the same conditionals, *improving generation quality reduces post-hoc watermarkability*.

## 1 INTRODUCTION

Existing language-model watermarks encode provenance in *what tokens are produced*. Red–green schemes bias the token distribution toward a keyed vocabulary partition (Kirchenbauer et al., 2023); distortion-free schemes fix the sampling randomness with a keyed pseudorandom sequence; and the watermarks designed for diffusion language models exploit the extra freedom those models expose — unresolved context, per-step sampling randomness, global sequence statistics, or the order in which positions are unmasked (Hong & No, 2026). Different as they are, every one of them leaves the verifier reading a *surface signature*: a hash of token identities, a frequency, a rank. We ask a different question.

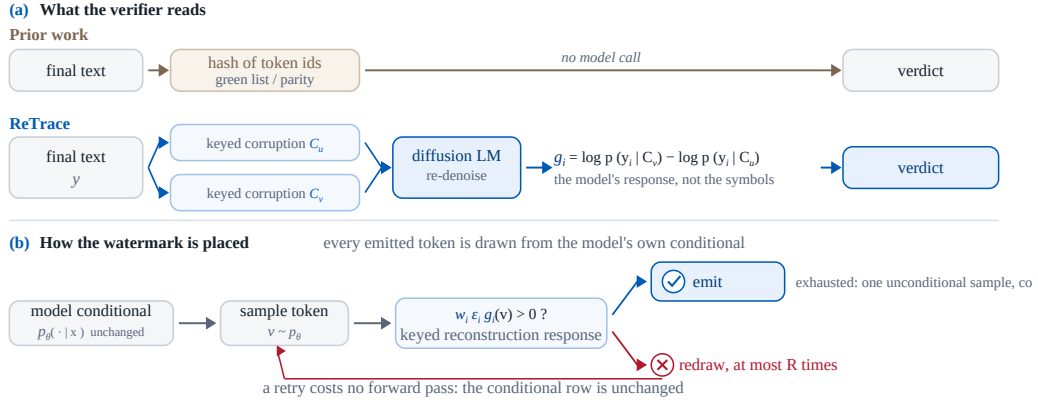
*Can a watermark be detected from how the model reconstructs a text, rather than from the tokens the text contains?*

A diffusion language model makes this question answerable in a way an autoregressive model does not. Its training objective is to recover masked tokens from arbitrary surviving context, so any finished text can be pushed *back* through the operator that produced it: corrupt it, re-denoise it, and observe the response. That response is a property of the text’s relationship to the model, not of its symbols, and it is available to anyone holding the model — but only interpretable by someone holding the key that chose the corruption.

We call the resulting scheme **ReTrace**, and its verifier is easiest to describe as a keyed question posed to every token of the text. The key hides two different parts of the text’s own context — say, the sentences nearest a span in one version and the farthest in the other — and the model scores the span’s tokens under both corruptions. Each token then *answers* the question “which context do you rely on more?”: the sign of

$$g_i(y_i) = \log p_{\theta}(y_i | C_i^1(y)) - \log p_{\theta}(y_i | C_i^0(y)). \quad (1)$$

The key also fixes, per position, which answer a watermarked text *should* give, and the verifier counts agreements. A text written without the key cannot know the questions, so each of its answers matches an independent hidden coin: agreements follow an exact  $\text{Binomial}(n, \frac{1}{2})$  law, and prove-



**Figure 1: Prior verifiers read token identities; ReTrace’s verifier runs the model. (a)** The finished text is corrupted twice under the key and re-denoised; the contrast between the reconstructions of each observed token is the evidence. **(b)** Embedding never leaves the model’s conditional: sample, check the keyed response, resample at most  $R$  times at no forward-pass cost, and commit an unconditional sample on exhaustion.

nance becomes an ordinary hypothesis test. No vocabulary partition, no token hash — the evidence lives in the model’s behaviour.

How does the watermark get the answers *into* the text? Not by editing tokens after generation: on a well-trained denoiser the median position admits exactly one fluent token, so there is nothing to edit toward, and the effect *strengthens* as generation improves (§4.3) — *better diffusion-model generation reduces post-hoc watermarkability*, ruling out an entire family of designs.

What survives is to never override the model at all. ReTrace decodes exactly as the base model would, with one added rule: at positions where the keyed question is genuinely open — both answers plausible, which the key can determine on its own — the model’s sampled token is checked against the desired answer, and *resampled from the same distribution* (at most  $R$  times, at no extra forward pass) until an agreeing token comes up; if none does, the natural sample is kept. Every committed token is therefore a sample the model itself proposed. A key-free *plausibility floor*  $p(v) \geq \kappa p_{\max}$  additionally refuses agreeing-but-unlikely tokens, capping what any single position can cost. The measured effect of both guards together is that watermarked perplexity sits *at or below* the matched non-watermarked control ( $0.74 \times - 0.87 \times$ ) at TPR@1% of 0.70–0.80 on 512-token documents, with the whole embedder costing  $1.28 \times$  a plain decode.

*ReTrace places the watermark in resampled tokens and reads it back from the model’s reconstruction response* (Figure 1).

In sum: a watermark read from the model’s response rather than its tokens, verifiable from text, model and key alone (§4); a counting statistic with a finite-sample exact binomial null under the ideal-PRF model, false-positive control verified at the *worst* of twenty keys (§4.2); an embedding that never leaves the model’s conditional and holds quality at or below its control (§4.4); a mechanism-level negative result — substitution is priced out, and better generation makes it worse (§4.3); and an evaluation that reports what each method pays, not only what it scores — generation budget and detection cost per row, with the advantages *and* the limitations of the design stated explicitly against them (§5).

## 2 RELATED WORK

**Diffusion language models.** Continuous diffusion (Ho et al., 2020) was carried to discrete data by structured corruption processes (Austin et al., 2021) and by score-ratio and masked-diffusion formulations (Lou et al., 2024; Sahoo et al., 2024; Shi et al., 2024; Ou et al., 2025). Instruction-following models at language-model scale followed — LLaDA and its successors (Nie et al., 2025; Zhu et al., 2025), Dream (Ye et al., 2025) — alongside industrial systems built on the same paradigm (Labs

et al., 2025). The property we exploit is the training objective itself: these models predict masked tokens from arbitrary surviving context, so they can be applied to a *finished* text and not only used to produce one.

Two further findings from this literature shape our design. First, practical dLLMs are *not* order-invariant even though an ideal conditional predictor would be, and the choice of unmasking order measurably changes what is generated (Kim et al., 2025). Second, decoding is in practice semi-autoregressive: blocks are filled in sequence and diffusion happens within a block (Arriola et al., 2025), and work on accelerating that loop shows that committing a confident token whose left context is still unresolved damages coherence (Wu et al., 2026; Wei et al., 2026; Ben-Hamu et al., 2025). Our embedder inherits both: it steers only *within* the reference block schedule, and it borrows the frontier-anchored commit rule rather than inventing a schedule of its own.

**Watermarks for autoregressive models.** The dominant family biases token probabilities. Kirchenbauer et al. (2023) seed a green/red vocabulary partition on the preceding token and add a logit bias; detection counts green tokens against a binomial null, and the scheme’s reliability and robustness have been studied extensively (Kirchenbauer et al., 2024; Zhao et al., 2023; 2024). A second family avoids distortion by fixing the sampling randomness with a keyed pseudorandom sequence, either through the Gumbel-max trick (Aaronson & Kirchner, 2023; Fu et al., 2024) or with explicit distortion-freeness and undetectability guarantees (Kuditipudi et al., 2024; Christ et al., 2024); others place the signature in model parameters rather than in sampling (Block et al., 2025). Surveys cover the space (Petitcolas et al., 1999; Liang et al., 2026), and the threat model is correspondingly well studied, from paraphrase attacks (Krishna et al., 2023) to extraction of the key itself (Jovanović et al., 2024). What every one of these has in common is the *verification* step: the detector computes a function of the token identities.

**Watermarks beyond left-to-right generation.** Because the green list is seeded by a prefix that order-agnostic decoding does not provide, transferring the autoregressive constructions to dLLMs requires care. Chen et al. (2025) give a watermark for order-agnostic models by biasing token selection in a way that does not assume a prefix. For diffusion language models specifically, Gloaguen et al. (2026) apply a red–green bias *in expectation* over unresolved context and additionally prefer tokens that make other positions green; Bagchi et al. (2025) use keyed Gumbel-max sampling at each denoising step while preserving the sampling distribution; Wu et al. (2025) and Raban et al. (2026) target the same setting with order-agnostic and efficiency-oriented constructions. Closest to our embedding channel is dgMARK (Hong & No, 2026), which leaves the learned conditionals untouched and instead prefers to unmask positions whose already-preferred token satisfies a keyed parity condition; that discipline is precisely why it is inexpensive, and we adopt it.

**What is new here.** All of the above differ in *where the watermark is placed* — token probabilities, sampling randomness, decoding order — while agreeing on *what the verifier reads*: a function of the emitted symbols. ReTrace changes the second. The verifier masks part of the finished text under the key, has the model reconstruct it, and compares the reconstructions; the evidence is the model’s response, which no token-level statistic can express. The price is equally clear and we report it in every table: those detectors need no model calls, ours needs a fixed number of sequence evaluations. To our knowledge no prior watermark for language models — autoregressive or diffusion — uses the generator itself as the verification oracle in this way.

### 3 BACKGROUND AND PROBLEM SETUP

**DLM generation.** A diffusion language model reverses a masking corruption: inference fills a masked region over  $T$  steps, committing the most confident positions each step, and practical systems decode semi-autoregressively — contiguous blocks left to right, diffusion within the active block (Arriola et al., 2025; Nie et al., 2025). Two consequences matter. The model predicts masked tokens from *arbitrary* surviving context, so it can be applied to a finished text and not only used to produce one; and each committed token is a sample from a conditional the verifier can reconstruct exactly, since that conditional sees only the finished prefix and masks.

**Watermarking problem.** An embedder holding a secret key  $K$  produces text whose provenance a verifier holding the same key can later establish from the text alone, against three requirements. *Detectability*: a test statistic separating watermarked from natural text at a controlled false positive rate. *Quality*: the watermarked distribution should cost little against the same generator without the key. *Verifiability without the trajectory*: detection may use the text, the model and the key, but nothing recorded at generation time. Prior schemes meet these by making the verifier read a function of the token identities — a green-list membership seeded by the preceding token (Kirchenbauer et al., 2023), a parity of the unmasking order (Hong & No, 2026), a keyed sampling sequence (Kuditipudi et al., 2024). This paper asks the verifier to read something else: how the model *reconstructs* the text under keyed corruption. The cost of that choice is a detector that must run the model, and we account for it in every comparison.

## 4 RETRACE

One requirement organises everything in this section. The verifier holds only the finished text, the model and the key — so whatever quantity the embedder steers toward during generation, the verifier must later be able to recompute *bit for bit* from the finished text. We first construct an observable with exactly that property (§4.1), then a statistic over it whose null distribution is known exactly rather than asymptotically (§4.2), then ask how generated text can be made to score highly on it. The obvious embedding fails for measurable reasons (§4.3), and what those measurements leave standing is the resampling embedder of §4.4.

### 4.1 THE OBSERVABLE

Let  $y$  be a generated span and  $K$  a secret key. For a block  $B$  of  $y$ , the key derives  $L$  *ablation patterns*  $D_1, \dots, D_L$  over the context preceding  $B$ , and a pair  $(u, v)$  among them. Writing  $\mathcal{C}_\ell$  for the context with  $B$ , everything after  $B$ , and  $D_\ell$  masked, the *challenge contrast* at position  $i \in B$  is

$$g_i(w) = \log p_\theta(w | \mathcal{C}_v) - \log p_\theta(w | \mathcal{C}_u), \quad w \in \mathcal{V}. \quad (2)$$

In words: how much more does the model expect  $w$  at position  $i$  when the key has hidden one part of the context than when it has hidden the other.

The shape of  $\mathcal{C}_\ell$  is dictated by the reproducibility requirement. Mid-generation, when the embedder consults Eq. (2), every block after  $B$  is still [MASK]; if the verifier later recomputed the contrast on the finished text without re-masking that tail, it would condition on tokens that did not exist when the embedder acted, and the two sides would be reading different functions. So  $\mathcal{C}_\ell$  masks  $B$  and everything after it, in every arm. Masking  $B$  itself buys a second property for free: the contrast becomes *invariant* to whatever is committed inside  $B$ , so one table, computed once, serves the block’s entire decoding and is reproduced identically from the finished text. The ablations, for the same reason, lie only in the region that is already final —  $D_\ell \subset \text{prompt} \cup B_{<b}$ , which the verifier has verbatim. Within that region the pair is chosen for signal: one pattern ablates the context nearest the block, the other the farthest, and because recency dominates masked-token prediction, the two arms disagree far more than two random ablations would, widening the dynamic range of  $g$ .

### 4.2 THE STATISTIC

The contrast turns each position into a signed answer; the statistic aggregates the answers so that chance is exactly quantifiable. Each position carries an independent keyed *orientation bit*  $\varepsilon_i \in \{\pm 1\}$  — the hidden coin fixing which sign of  $g_i$  counts as agreement — and a target  $w_i \in \{\pm 1\}$  carrying the message bit ( $w_i \equiv +1$  when only provenance is claimed). Aggregation is a plain count of agreements,

$$m_i = \mathbf{1}[w_i \varepsilon_i g_i(y_i) > 0], \quad T = \sum_{i \in P} m_i, \quad (3)$$

and a count rather than an average deliberately: the per-position contrast is heavy-tailed, so an average is dominated by its largest values and sign steering has no leverage on it, whereas a count gives every position exactly one unit.

Why is chance exactly quantifiable? Because for a text produced without the key, each  $m_i$  is a comparison against a coin that was never consulted:

**Proposition 1** (Exact null). *Let the challenge patterns, the per-position pair selection, the carrier set, the group assignment, the tie coins and the orientation bits each be derived from the key under separate PRF domains. Fix a text  $y$  independent of the key, and condition on the carrier set, the selected pairs, and the challenge magnitudes — all of which are functions of  $y$  and the non-orientation domains only. Under the ideal-PRF model the orientation bits are then independent Rademacher variables independent of everything conditioned on, so the  $m_i$  are i.i.d. Bernoulli( $\frac{1}{2}$ ) and  $T \sim \text{Binomial}(|P|, \frac{1}{2})$  exactly.*

The result needs no calibration corpus and no model-specific null estimate, but its hypotheses have to be engineered, not assumed. The separate PRF domains let the conditioning go through: pattern choice and pair selection are themselves keyed, and sharing randomness with the orientation bits would leak the coin (pair selection uses only the swap-invariant  $S_i$  of Eq. (7), so it is orientation-symmetric). Exact ties  $g_i(y_i) = 0$  are resolved by a further keyed coin rather than scored as misses, which would bias  $T$  off the binomial. And presence is tested on its own pool with  $w_i \equiv +1$ : summing agreement across payload groups cancels — a balanced message leaves a perfect watermark at the chance value  $|P|/2$  — so claimed-message verification aligns each group to its bit and payload decoding thresholds each group separately.

#### 4.3 WHY SUBSTITUTION CANNOT PAY

With the observable and the statistic fixed, the embedder’s task sounds elementary: make the committed tokens answer with the key. The natural first design does it by *substitution* — generate normally, then at keyed positions replace the model’s token with an alternative whose contrast points the right way, subject to a per-token fluency budget  $\tau$  (admissible iff  $p(w)/p(w^*) \geq e^{-\tau}$ ). Its measured failure dictates the final design.

The first obstacle is that there is usually nothing to substitute. A well-trained denoiser’s conditionals are sharp: across 36 configurations spanning three carrier-selection rules, three budgets and two guidance weights, the *median* keyed position admits exactly one token at  $\tau = 3$  ( $p(w)/p(w^*) \geq 0.05$ ), with no truncation of the candidate slate. The best configuration inside a  $1.35\times$  perplexity budget reaches only TPR@1% = 0.10, against 0.86 for locally reproduced dgMARK at  $1.23\times$ .

Sharper still, the obstacle grows with model quality. When a confidence-ranking defect in our sampler was repaired — draft perplexity moving from 22.2 to 9.5 — the denoiser’s entropy halved ( $0.655 \rightarrow 0.319$ ) and the admissible sets shrank by 34–60% ( $|A(3)|$ :  $2.5 \rightarrow 1.6$ ;  $|A(6)|$ :  $14.1 \rightarrow 5.6$ ). Substitution buys evidence with fluency at a rate set by the sharpness of the model’s conditionals, and *improving a diffusion language model’s generation quality reduces its post-hoc watermarkability*, because both quantities are governed by the same sharpness. Nor does moving the substitution to generation time escape this: a guidance table valid during generation needs every non-carrier token final first, and the two-phase schedule that forces costs  $1.98\times$  perplexity *before any watermark* — more than dgMARK’s entire cost.

What these measurements leave standing is a design constraint rather than a tuning problem: an embedder must never override the model’s choice — it may only choose *among* tokens the model itself proposes. The next subsection builds exactly that embedder.

#### 4.4 EMBEDDING BY RESPONSE-GUIDED RESAMPLING

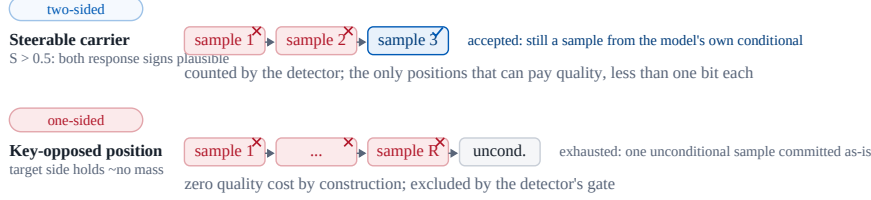
Rejection resampling is the embedder that satisfies this constraint. Generation follows the reference schedule of the base model — blocks in order, diffusion within a block, positions committed in plain confidence order (Figure 2) — and Algorithms 1 and 2 put the embedder beside the reference sampler it wraps, line for line: the two differ only in the *blue* lines — no logit is edited, no commitment is reordered, and deleting the blue lines recovers the base model’s decoder exactly. At a carrier position the committed token is chosen by rejection sampling from the model’s own conditional:

$$v^{(1)}, v^{(2)}, \dots \sim p_\theta(\cdot | x), \quad \text{commit the first } v^{(r)} \text{ with } w_i \varepsilon_i g_i(v^{(r)}) > 0, \quad r \leq R, \quad (4)$$

and if all  $R$  guided proposals are rejected, one further unconditional sample is committed as-is. Acceptance may additionally require a *plausibility floor*  $p_i(v) \geq \kappa p_i^{\max}$ , which caps any carrier’s cost at  $-\log \kappa$  beyond the model’s own argmax; the floor is embedder-only and key-free — the detector reads nothing but the committed token — so the null and the carrier rule are untouched. A

| Algorithm 1 LLaDA reference sampler                                                                                                                                                                                                                                                                           | Algorithm 2 ReTrace in LLaDA (blue = ours)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Input:</b> prompt<br><b>Output:</b> text $x$<br>1: <b>for</b> each block $b$ in order <b>do</b><br>2:   —<br>3: <b>while</b> $b$ has masks <b>do</b><br>4:     select highest-confidence masked $i$<br>5:     sample $v \sim p_\theta(\cdot   x)$<br>6:     —<br>7:     commit $v$ at $i$<br>(no verifier) | <b>Input:</b> prompt, key $K$ , budget $R$ , floor $\kappa$<br><b>Output:</b> text $x$<br>1: <b>for</b> each block $b$ in order <b>do</b><br>2:   challenge table for $b$ : masked fwds<br>$\rightarrow g_i$ , gate $S_i$ , signs $w_i \varepsilon_i$ (§4.1)<br>3: <b>while</b> $b$ has masks <b>do</b><br>4:     select highest-confidence masked $i$<br>5:     sample $v \sim p_\theta(\cdot   x)$<br>6: <b>if</b> $S_i > s_{\min}$ : resample ( $\leq R$ ) until<br>$w_i \varepsilon_i g_i(v) > 0 \wedge p_i(v) \geq \kappa p_i^{\max}$ ;<br>on exhaustion, one unconditional sample (Eq. 4)<br>7:     commit $v$ at $i$<br><b>Verify:</b> from $x$ : rebuild line-2 tables under $K$ ,<br>count $\mathbf{1}[w_i \varepsilon_i g_i(x_i) > 0] \rightarrow \text{Binom}(n, \frac{1}{2})$ (Eq. 3) |

Two kinds of position, one rule: sample from the model until the keyed response accepts, at most  $R$  times.



**Verification** from the finished text alone, with no access to the trajectory or the draw sequence



**Figure 2: One rule, two regimes.** A two-sided carrier accepts within a few resamples — still a sample from the model’s conditional, and the only kind of position that pays quality. A key-opposed position rejects every proposal and commits the natural sample at zero cost; the gate excludes it. The verifier re-masks under the key and counts gated indicators against an exact binomial null, with no access to the trajectory.

retry costs no forward pass — the conditional row is unchanged — so  $R$  trades detection strength against arithmetic alone, and the floor keeps that trade from spending the quality budget. Writing  $A_i$  for the acceptance set and  $q_i$  for its mass, the per-position hit probability and committed-token distribution are

$$q_i = \Pr_{v \sim p_i} [w_i \varepsilon_i g_i(v) > 0], \quad \Pr[m_i = 1] = 1 - (1 - q_i)^{R+1}, \quad (5)$$

$$v_i \sim (1 - (1 - q_i)^R) p_i(\cdot | A_i) + (1 - q_i)^R p_i. \quad (6)$$

This *does* condition the sampling distribution on the acceptance predicate and is not claimed to be distortion-free; what makes it cheap is that the support never leaves the model’s own conditional, and that the positions where the predicate binds hardest are exactly the positions where it cannot be satisfied at all — which fall through to the natural sample.

One question remains: which positions should the detector count? Not all of them — the acceptance mass  $q_i$  is bimodal in practice (quartiles 0.02/0.60/0.98), meaning at most positions the model’s plausible tokens all answer the challenge the same way. Where that answer opposes the key, retries cannot help, and counting the position turns a coin flip into a certain miss. Positions are therefore gated by the two-sided mass

$$S_i = 2 \min(q_i^+, q_i^-), \quad q_i^\pm = \Pr_{v \sim p_i^{\text{base}}} [\pm g_i(v) > 0], \quad (7)$$

| Method                                                                          | Setting                          | PPL ↓ | Ratio ↓ | TPR at FPR ↑ |      |      | Gen. fwd. | Det. fwd. |
|---------------------------------------------------------------------------------|----------------------------------|-------|---------|--------------|------|------|-----------|-----------|
|                                                                                 |                                  |       |         | 5%           | 1%   | 0.1% |           |           |
| Budget: one decode per document (Gen. fwd. = tokens generated)                  |                                  |       |         |              |      |      |           |           |
| dgMARK (Hong & No, 2026)                                                        | $k = 1$ , 256 tok                | 15.40 | 1.55×   | 0.88         | 0.74 | 0.62 | 256       | 0         |
| KGW (Kirchenbauer et al., 2023)                                                 | $\delta = 1$ , 256 tok           | 11.96 | 1.03×   | 0.97         | 0.93 | 0.73 | 256       | 0         |
| KGW (Kirchenbauer et al., 2023)                                                 | $\delta = 3$ , 256 tok           | 14.05 | 1.21×   | 1.00         | 1.00 | 1.00 | 256       | 0         |
| ReTrace V1                                                                      | best under 1.35× ppl             | 30.5  | 1.17×   | 0.10         | 0.10 | 0.00 | 256       | 9         |
| Budget: 1.28× decode — ReTrace V3 (decode + one challenge-table pass per block) |                                  |       |         |              |      |      |           |           |
| ReTrace V3                                                                      | $R=1$ , 512 tok                  | 11.0  | 0.97×   | 0.57         | 0.43 | 0.20 | 656       | 144       |
| ReTrace V3                                                                      | $R=2$ , 512 tok, held-out        | —     | 1.14×   | 0.83         | 0.73 | 0.63 | 656       | 144       |
| ReTrace V3                                                                      | $R=4$ , 512 tok, held-out        | —     | 1.40×   | 0.90         | 0.83 | 0.77 | 656       | 144       |
| ReTrace V3                                                                      | $R=16$ , $\kappa=0.05$ , 512 tok | —     | 0.74×   | 0.90         | 0.80 | 0.60 | 656       | 144       |
| ReTrace V3                                                                      | $R=8$ , $\kappa=0.1$ , 512 tok   | 5.67  | 0.87×   | 0.80         | 0.70 | 0.60 | 656       | 144       |
| ReTrace V3                                                                      | $R=8$ , $\kappa=0.1$ , 1024 tok  | —     | 0.83×   | 0.81         | 0.75 | 0.69 | 1312      | 288       |
| Budget: ~4× decode (one-step lookahead per candidate)                           |                                  |       |         |              |      |      |           |           |
| dgMARK (Hong & No, 2026)                                                        | +3-beam, 256 tok                 | 12.23 | 1.23×   | 0.92         | 0.86 | 0.82 | ~1024     | 0         |

**Table 1: Detectability against the quality and compute actually paid, grouped by generation budget.** One machine, checkpoint and prompt protocol. *Ratio* is against a non-watermarked control run under the same decoding regime (control PPL: block decoding 9.94, left-to-right 11.61,  $T=0.8$  full-vocab 26.21; ReTrace: paired matched-scheduler control per sub-group); control arms score  $\text{TPR} \leq 0.03$  at the 5% threshold and 0.00 below,  $|z| \leq 0.25$ . *Gen. fwd.* counts model forwards spent producing the text; ReTrace’s tier costs  $1.28\times$  a plain decode for its per-block challenge tables, and dgMARK 3-beam sits alone at  $\sim 4\times$  for its lookahead. *Det. fwd.* counts masked evaluations per detection — zero for token-hash detectors, fixed for ReTrace. KGW is scored on distinct bigrams (37% FPR at  $\delta=0$  otherwise); its left-to-right regime itself costs  $1.17\times$  against block decoding. ReTrace prompt sets are disjoint per sub-group with per-document keys,  $n=30$  per 512-token arm ( $n=20$  for  $R=16$ ,  $n=16$  at 1024 tokens; quality pairing on valid documents).

and only carriers with  $S_i > s_{\min}$  are counted. The gate respects both earlier requirements at once: it is computed from the block-masked conditional, so the verifier reconstructs the identical carrier set from the finished text (§4.1); and it is invariant under swapping the pair’s arms and never sees the key’s direction or the message, so conditioning on it leaves the null of Proposition 1 intact. The challenge pair itself is chosen per position among the  $\binom{L}{2}$  keyed patterns by the same swap-invariant  $S_i$ , which raises the mean two-sided mass from 0.28 (a fixed near/far pair) to 0.45.

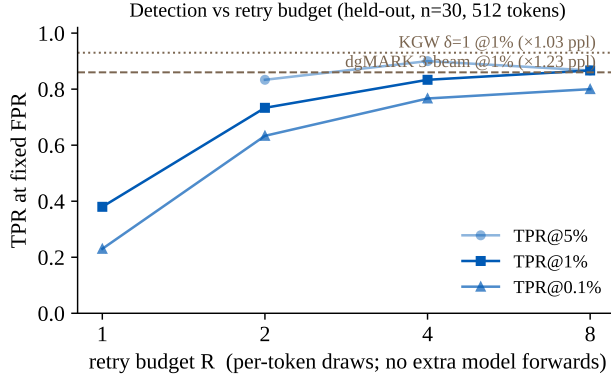
The loop is now closed: every quantity in Algorithm 2’s blue lines is a function of text, model and key alone (§4.1), the count they feed carries an exact null (§4.2), and the only freedom the embedder exercises — choosing among the model’s own proposals — is the one §4.3 showed a well-trained denoiser leaves open.

## 5 EXPERIMENTS

One TITAN RTX, GSAI-ML/LLaDA-8B-Instruct in fp16; C4 realnewslike prompts truncated to 300 characters (dgMARK’s protocol); 256–512 tokens at steps = gen\_length. Quality is GPT-2-large perplexity against a control *under the same decoding regime* — for ReTrace, the same generator with steering disabled, so the pairing isolates the watermark. Detectability is TPR at fixed FPR, never raw  $z$ ; development prompts are never reused, and confirmatory runs use disjoint corpus offsets with per-document keys  $K_d = \text{HMAC}(K, \text{nonce}_d)$ . Compute is a reported axis, not a footnote: generation forwards appear per row, and the budget-matched comparator for ReTrace ( $1.28\times$  a plain decode, from one challenge-table pass per block) is dgMARK  $k=1$  — its 3-beam variant spends roughly four decodes per document on lookahead and is flagged as such.

### 5.1 MAIN RESULTS AND ANALYSIS

**At low retry budgets the watermark costs nothing — measured against the right control.** Paired against a control from the *same* generator with steering disabled, the watermark’s  $\Delta\text{NLL}$  median is  $-0.027$  at  $R=1$  and  $-0.037$  at  $R=2$  (ratios  $0.97\times$ ,  $0.93\times$ ;  $n=30$ ), replicated on a disjoint held-out set at  $0.980\times$  ( $n=40$ ). The control matters: pairing against a separately-run reference generator with a different schedule overstates the cost roughly twofold, because the pairing then measures the scheduler difference, not the watermark. The mechanism says the near-zero cost is not



**Figure 3: Retries buy detection with arithmetic, but spend the quality budget.**  $n=30$  held-out 512-token documents, per-document keys, control at zero false positives; paired quality  $1.14 \times / 1.40 \times / 1.61 \times$  at  $R=2/4/8$ . Reference lines are this paper’s local baseline reproductions.

luck: a position whose plausible tokens all answer one way commits the natural sample whichever way the hidden coin landed, so only genuinely two-sided carriers pay, each less than one bit.

**Retries buy detection without compute — but, unguarded, they do spend quality.** On held-out prompts ( $n=30$ , 512 tokens), TPR@1% climbs  $0.73 \rightarrow 0.83 \rightarrow 0.87$  across  $R=2/4/8$  while a retry costs no forward pass (Figure 3); the paired quality on the same run is  $1.14 \times$ ,  $1.40 \times$  and  $1.61 \times$ . The reason both things are true at once: resampling is free in compute, but each extra retry makes it likelier that a *less probable* agreeing token is committed, and that likelihood loss is the quality cost. The operating points that survive comparison are therefore the low- $R$  ones: at  $R=4$  the detection is level with locally reproduced dgMARK 3-beam ( $0.86$ ) but at higher quality cost ( $1.40 \times$  against  $1.23 \times$ ), so dgMARK dominates that point, whereas nothing in the table reaches the  $R \leq 2$  region’s quality at any detection strength. Length is the other free axis: at  $R=1$ , doubling the document from 256 to 512 tokens lifts 1%-TPR from 0.25 to 0.40 purely by growing the presence count, the same length–power trade dgMARK reports.

**A plausibility floor decouples  $R$  from the quality budget.** The floor refuses exactly the tokens the previous paragraph identified as the cost — agreeing but unlikely ones. Requiring  $p(v) \geq \kappa p_{\max}$  at acceptance caps each carrier’s cost at  $-\log \kappa$ , and at  $\kappa=0.1$  it converts the  $R=8$  point from  $1.61 \times$  to  $0.87 \times$  while keeping TPR@1% = 0.70 (the fraction of carriers answering with the key drops only from 0.711 to 0.620; the control arm stays at the chance level of 0.501). Repetition sits at parity with the control (0.33 against 0.36 on decoded text under the shared validity rule). Sub-1.0 ratios deserve suspicion rather than celebration — the floor tilts toward higher-probability tokens, which perplexity rewards — which is why repetition is reported next to the ratio rather than instead of it. Doubling the document to 1024 tokens moves the same configuration to 0.75 at 1% and 0.69 at 0.1% (rate 0.635, ratio  $0.83 \times$ ), with the caveats stated rather than averaged away: repetition opens a  $+0.06$  gap over the control at this length, and only half the documents clear the validity rule for quality pairing. Calibrated and weighted detector variants were also evaluated on identical caches and buy at most 0.02 at 1% — the count is essentially optimal at these rates, so remaining power must come from the acceptance rate and the carrier count, not the statistic.

**The simplest detector is the right one.** On identical held-out caches the gated count, the gated likelihood ratio and an all-position likelihood ratio are indistinguishable (0.38 at 1% each at  $R=1$ ), ungated counting is strictly worse (0.33), and every variant holds its null on the control arm. Likelihood weighting buys nothing here because its extra evidence — one-sided positions failing *visibly* — only materialises when the embedder has tried and been refused, which takes retries; at the operating points that matter the deployed detector is the gated indicator with its exact binomial null and no Monte-Carlo machinery.

| Nominal $\alpha$ | Mean FPR | Worst-key FPR | Verdict                 |
|------------------|----------|---------------|-------------------------|
| 0.10             | 0.0121   | 0.0583        | valid ( $\leq \alpha$ ) |
| 0.05             | 0.0037   | 0.0250        | valid ( $\leq \alpha$ ) |
| 0.01             | 0.0000   | 0.0000        | valid ( $\leq \alpha$ ) |

**Table 2: False positive control holds per key, worst key included.** 20 keys  $\times$  120 non-watermarked generations under the deployed rule. Deployment uses one key across documents, so the per-key guarantee is the one that matters; an earlier Gaussian-tail version measured 0.145 at nominal 0.10 and was withdrawn.

**False positive control holds at the worst key.** Twenty keys  $\times$  120 non-watermarked generations give 2400 null trials under the deployed decision rule: empirical FPR is at or below nominal at every level *per key*, the worst key included (0.058 at  $\alpha=0.10$ , 0.025 at 0.05, 0.000 at 0.01; Table 2), and the matched control arms of every later run sat at zero false positives; the exact construction is what lets the guarantee be stated per document rather than on average.

**Robustness is the weak axis, and the mechanism says why.** Same-model re-denoising of 10% of positions collapses  $R=1$  detection from 0.35 to 0.05 at 1% (deletion likewise), while the attacked control stays at  $\alpha$  — the attack erases evidence rather than triggering the detector. The fragility is structural: every block’s challenges condition on all earlier blocks, so one edit perturbs the contrast of every later position. The honest current statement is that token-hash watermarks keep an advantage under heavy post-editing.

**Advantages, stated as claims the tables support.** (i) *Quality below the control*: with the plausibility floor, ReTrace is the only method in Table 1 operating under  $1.0 \times (0.74 \times -0.87 \times)$  at nontrivial detection — every baseline pays quality for strength. (ii) *An exact, per-document null*: the decision threshold is  $\text{Binomial}(n, \frac{1}{2})$ , not an asymptotic  $z$  or an empirically calibrated cutoff, and it held at the worst of twenty keys (Table 2). (iii) *The sample never leaves the model*: no logit is edited, no vocabulary is partitioned, no commitment is reordered (Algorithm 2), so failure modes are capped — a key-opposed position falls through to the natural sample at zero cost. (iv) *Near-plain generation budget*:  $1.28 \times$  a reference decode, against  $\sim 4 \times$  for the strongest dgMARK setting.

**Limitations, equally explicit.** (i) *Detection needs the model*: 144 masked evaluations per 512-token document ( $L+1 = 9$  per block), batchable and seconds on a modern accelerator, but never the hash detectors’ zero — this prices a verification API, not a mass-scanning filter, and the method is proposed only for the former. (ii) *Post-editing robustness is the weak axis*: the collapse measured above is structural, and token-hash schemes keep a real advantage under heavy edits. (iii) *Carrier supply depends on model entropy*: sharp conditionals yield few two-sided positions, and the same mechanism that priced out substitution (§4.3) caps how many carriers a well-trained model offers. (iv) *Detection strength trails the strongest baselines* (0.80 at 1% against 3-beam dgMARK’s 0.86, KGW’s 0.93) — the case for ReTrace is the joint operating point, not any single axis.

## 6 CONCLUSION

ReTrace turns the fact that a diffusion language model can be *applied* to a finished text into a verification channel: corrupt the text under a secret key, re-denoise, and read whether the reconstruction response agrees with a keyed orientation. Embedding is a keyed rejection-resampling that never leaves the model’s own conditional; with a key-free plausibility floor on acceptance it holds sub-control perplexity even at high retry budgets —  $\text{TPR}@1\% = 0.70$  at  $0.87 \times$  with repetition at parity ( $R=8$ ,  $\kappa=0.1$ , held-out). The frontier position is honest rather than dominant — locally reproduced dgMARK (0.86 at  $1.23 \times$ ) still leads on detection, but nothing in the comparison reaches this quality region at any strength — and the null is exact per document, holding empirically at the worst of twenty keys.

What separates ReTrace from the watermarks it is compared against is the object verified: token hashes read symbols, ReTrace reads the model’s response, and §5 keeps the explicit ledger of what that buys and costs. The open item on that ledger is post-editing robustness; hardening the response channel against edits is the next thing this line of work must address.

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